



TM2 Review Guide for Test 2 on 02.26

Completion requirements

Test format. Test 1 will be administered at the start of class. You will have 20 minutes to complete the test. It will consist of short answer and simple problem solving. You may occasionally encounter multiple choice questions. After the test, we will introduce the first lab, and you will begin working on it.

Remember that our assigned room has changed: We will be in Olin 107 on test days.

Core topics related to data prep:

- Make sure you have completed all of the **assigned reading** (see links elsewhere on Moodle).
 - Note that I will often include a section called **Additional Resources or For Further Study**. These are primarily for reference or for further study of topics that interest you. These will *not* be covered on a quiz, test, or exam *unless a homework question or review guide specifically mentions one of these resources*.
- What is a **scatter graph** (a.k.a. **XY graph**)? How does it differ from a line graph or bar chart?
- What is a **histogram**? How does a histogram differ from a scatter graph, line graph, and bar chart?
- Give a high-level overview of the process of data preparation for a data analytics project in industry? [EMC2.3]
- Suppose you were adapting the content in Section 2.3 of our EMC textbook for students who were preparing for a term project in DSC 340, rather than an industry analytics project. What changes would you make for a smaller project such as this?
- What is **web scraping**?
- Explain why a team should **catalog the data sources** to which it has access.
- Be prepared to describe several of the major data conditioning tasks, such as:
 - **flattening** data
 - **structuring** data
 - **cleaning** data
 - **transforming** data
 - **scaling**
 - **encoding** categorical data
 - (You should also be prepared to suggest suitable tools for these tasks, if we discussed them in class or if it is covered in the reading.)

- I may give you a simple data set with a few rows and a couple of features, and some standard statistics, such as mean, variance, standard deviation, etc., and ask you to:
 - Scale the data in the same way that sklearn's **MinMaxScaler** does this.
 - Scale the data in the same way that sklearn's **StandardScaler** does.
 - (Note that each **column** is scaled separately.)
 - You should not need a calculator for this. Either the numbers will work out nicely for pencil and paper math, or you will be allowed to express the result as an expression that is not in simplest form. (The instructions will clarify which.) Of course, you need to know both formulas to answer these questions (see slides).
- Make sure you know the difference between **scaling columns** and **normalizing rows**. Note: we only talked about normalizing rows for text embeddings. Scaling the columns is much more common.
- Describe specific actions that you can take to survey and visualize your data in preparation for model planning and building. (I will be looking for specific tools and tasks, not vague pabulum about why visualization is important.)
- What is **granularity** in the context of a data analytics project? Give a specific example.
- What is **dirty data**? Be prepared to provide several specific examples.
- What are some actions that you can take for cleaning your data?
- What is **categorical data** and how does it differ from **numeric data**? Give example of each.
- Name and briefly describe several methods that you could use to deal with **missing data** (e.g., ignore it, mean imputation, etc.).

Core topics related to text embeddings:

- What is **feature space**?
- I may give you a simple dataset with a few columns (features) and ask you the number of **dimensions** of the feature space. Hint: it is the number of columns after you have done any transformations such as adding or dropping columns, one-hot encoding, etc.
- What is a text or word **embedding**?
- Be prepared to describe some applications of text embeddings.
- What is **semantic distance**? Given three words or phrases, be prepared to tell which two have the greatest semantic distance and why.
- Make sure you **download the [Visualizing Text Embedding.ipynb](#)** files in the [Embeddings Code](#) folder on Moodle and read through that. It has some required reading.** Afterwards, make sure you know how to:
 - Compute the **dot product** (like the slide in class).
 - Compute the **magnitude** (length) of a vector.

- **Normalize** a vector.
- Find the angle between two vectors. (Note: I will either give you a table for the arccos or you can just leave it written as arccos of some number.)
- Compute the **cosine similarity** and **cosine distance**.
- What is a **unit vector**?
- Make sure you know that if we are comparing unit vectors, the bigger the dot product, the smaller the angle, and hence the more similar the data points (and vice versa).
- If the dot product = 0, what can you say about the angle between the vectors?
- Why do we use projections such as PCA or t-SNE to visualize high-dimensional data?
- What is a **recommender system**?

As usual, if you have any questions, please contact me ASAP. Good luck! -Prof. Allen

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